
Explainable Prediction of Medical Codes from Clinical Text

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Introduction

- Clinical notes
 - Free text narratives generated by clinicians during patient encounters
- International Classification of Diseases (ICD) codes
 - Metadata codes that accompany clinical notes
 - Standardized way of indicating diagnoses and procedures
 - Multiple use cases (billing, predictive modelling)

HISTORY

47 year old male with mid-abdominal epigastric pain¹, associated with severe nausea & vomiting; unable to keep down any food or liquid. Pain has become "severe" and constant.

Has had an estimated 13 pound weight loss over the past month.

Patient reports eating 12 sausages at the Sunday church breakfast five days ago which he believes initiated his symptoms.

Patient admits to a history of alcohol dependence². Consuming 5 – 6 beers per day now, down from 10 – 12 per day 6 months ago. States that he has nausea and sweating with "the shakes" when he does not drink.

EXAM

VS: T 99.8°F, otherwise normal.

Mild jaundice noted.

Abdomen distended and tender across upper abdomen³. Guarding is present. Bowel sounds diminished in all four quadrants.

Oral mucosa dry, chapped lips, decreased skin turgor

ASSESSMENT AND PLAN

Dehydration and suspected acute pancreatitis.

Admit to the hospital. Orders written and sent to on-call hospitalist.

1L IV NS started in office. Blood drawn for labs.

Recommend behavioral health counseling for substance abuse assessment and possible treatment.

Patient's wife notified of plan; she will transport to hospital by private vehicle.

ICD-9-CM DIAGNOSIS CODES

789.06 Abdominal pain, epigastric

789.60 Abdominal tenderness, unspecified site

782.4 Jaundice NOS

276.51 Dehydration

303.90 Other and unspecified alcohol dependence, unspecified

ICD-10-CM DIAGNOSIS CODES

R10.13 Epigastric pain

R10.819 Abdominal tenderness, unspecified site

R17 Unspecified jaundice

E86.0 Dehydration

F10.20 Alcohol dependence, uncomplicated

Motivation

- Limitation of ICD code annotation
 - Labor intensive
 - Coders read between 720 and 1,440 pages of Clinical Documentation per day²
 - Prone to error
 - Lack of connection bridging text and code
- Challenges+complexity of automatic coding
 - Features of clinical text
 - Irrelevant information, misspellings, non-standard abbreviations, large medical vocabulary
 - Label space

Summary of Contributions

- Convolutional Attention for Multi-Label classification (CAML)
 - Attentional convolutional network
 - CNN + per-label attention mechanism
 - Important information correlated with a code's presence may be in short snippets anywhere in the document
 - Decision support setting (clinical domain)
- Accuracy Evaluation
 - Used open-access MIMIC datasets
 - Outperformed previously proposed approaches
- Interpretability Evaluation
 - Physician rated the informativeness of a set of automatically generated explanations

Related Work: Automatic ICD Coding

- Structured text v. unstructured text
- Datasets
 - Subset of the full ICD label space (Wang et al., 2016)
 - Subset of medical scenarios (Zhang et al., 2017)
 - Death certificates in English + French (Névéal et al., 2017)
 - Private datasets (Subotin and Davis, 2016)
- Approaches
 - Recurrent networks with HA-GRU to classify ICD9 diagnosis codes (Baumel et al. (2018)
 - Character-aware LSTMs to generate sentence representations (Shi et al. 2017)
 - Memory networks to predict top-50 and top-100 Codes (Prakash et al. 2017)
 - Grounded Recurrent Neural Network to predict the presence of individual labels (Vani et al., 2017)

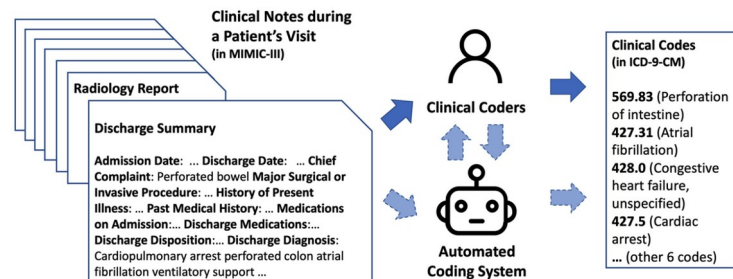


Image reference: <https://www.nature.com/articles/s41746-022-00705-7.pdf>

The CAML architecture yields stronger results across all experimental conditions.

Related Work: Explainable Text Classification

- Model “rationales” through a latent variable (Lei et al., 2016)
 - Tag each word as relevant to the document label
- Compute salience of individual words (Li et al., 2016)
- Use attention to highlight salient features of the text (Rush et al., 2015; Rocktäschel et al., 2016)
 - Lack interoperability evaluations of features selected by the attention mechanism

Related Work: Attentional Convolution for NLP

- CNNs have been applied to tasks such as sentiment classification (Kim, 2014) and language modeling
- Convolution and attention
 - Allamanis et al., 2016, Yin et al., 2016, dos Santos et al., 2016
 - Yin and Schütze, 2017
- Most relevant work (Yang et al. 2016 and Allamanis et al. 2016)
 - Use context vectors to compute attention over specific locations in the text
- Differentiation
 - Goal is to select locations in a document which are most important for predicting specific labels
 - Compute separate attention weights for each label in the label space

ICD Code Prediction Problem

A 12-year old girl with known hyperagglutinability, presented to the emergency department with a 2-week history of headaches and facial weakness. Neurologic examination indicated sensorineural hearing loss on the right side with Weber's test lateralizing to the left, and the Rinne's test demonstrating bone conduction greater than air conduction on the right. Magnetic resonance imaging of the head revealed severe structural defects of the right petrous temporal bone. No indication of cerebral infarction.

Clinical Text



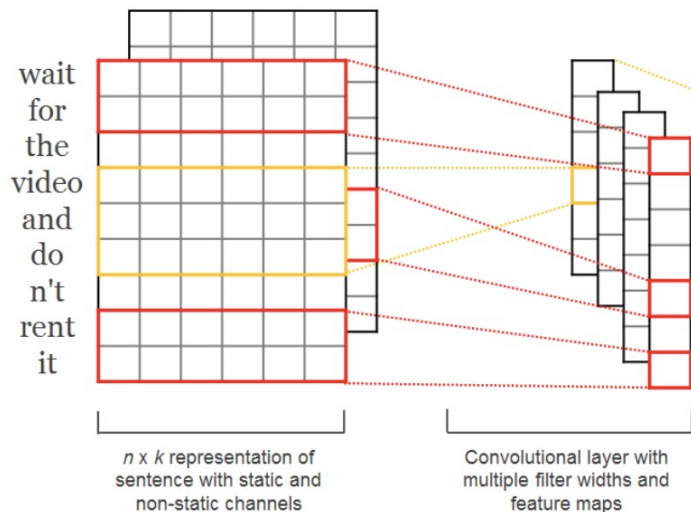
Diagnosis	ICD-9	ICD-10
Cervical Sprain, initial encounter	847.0	S13.4xxA
Thoracic Sprain, initial encounter	847.1	S23.3xxA
Lumbar Sprain, initial encounter	847.2	S33.5xxA
Cervical Degenerative Disc Disease	722.4	M50
Thoracic Degenerative Disc Disease	722.51	M51
Lumbar Degenerative Disc Disease	722.52	M51.2

ICD Codes

Multi-Label Classification Problem

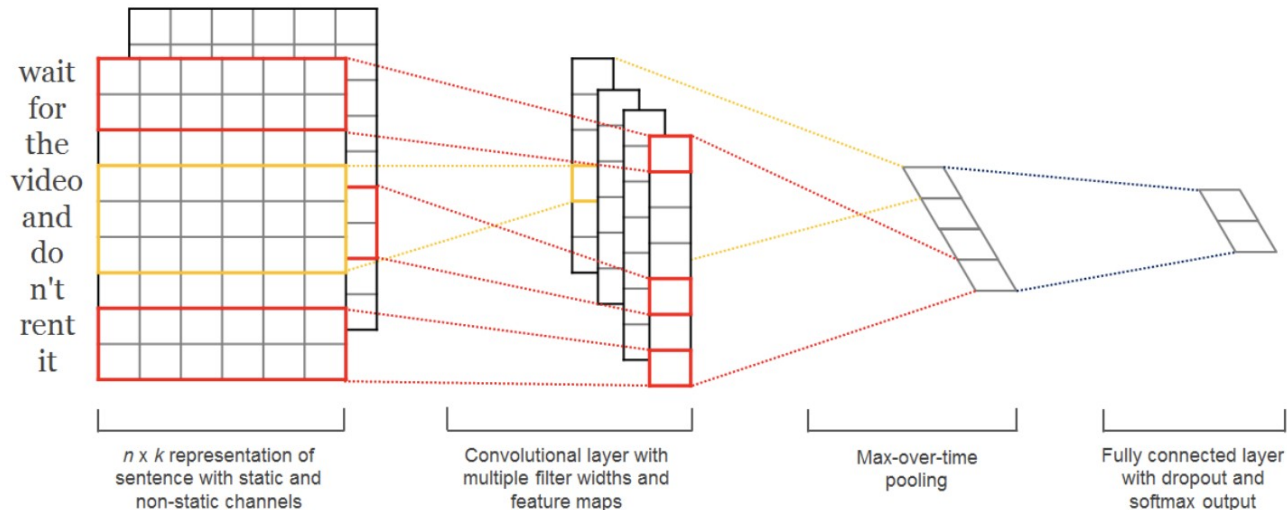
- “Match” snippets of clinical text to a label
- Use code descriptions to help make text snippets relevant to the predicted codes (esp. for labels with little data)

Typical Convolutional N-Gram Model



- Word embeddings
- Convolutional filter of "N" consecutive words (N-gram)
- Each convolutional filter "detects" certain "word snippet" motifs

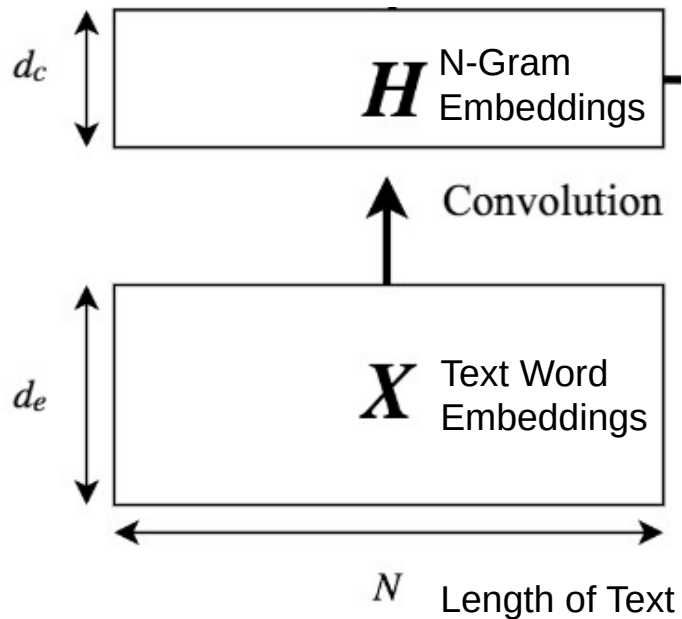
Typical Convolutional N-Gram Model



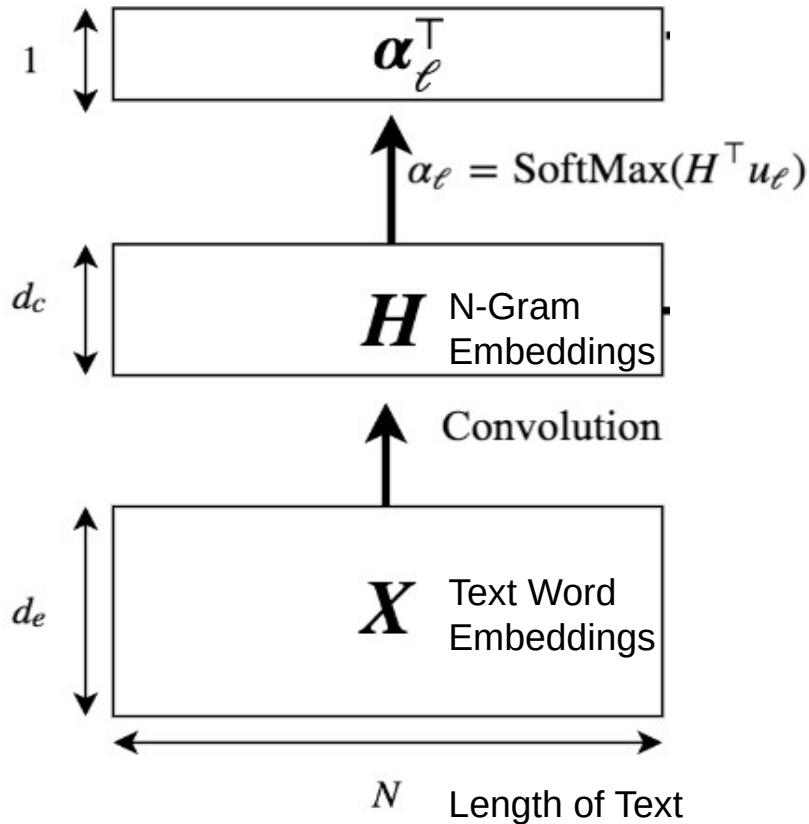
Max pooling → Was the “N-gram” detected at all in this piece of text?

- Word embeddings
- Convolutional filter of “N” consecutive words (N-gram)
- Each convolutional filter “detects” certain “word snippet” motifs

Convolutional Attention Model



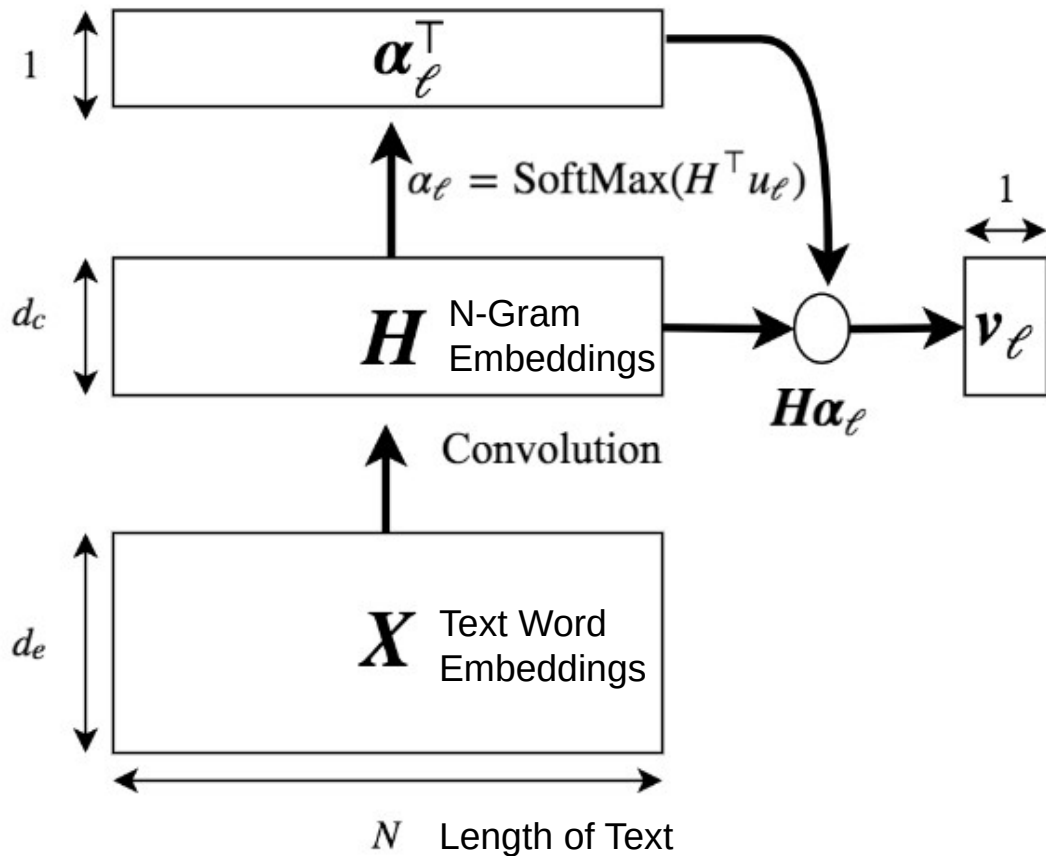
Convolutional Attention Model



$$\alpha_\ell = \text{SoftMax}(H^\top u_\ell),$$

Weight representing
“match” between n-grams in document (H) and label embedding

Convolutional Attention Model



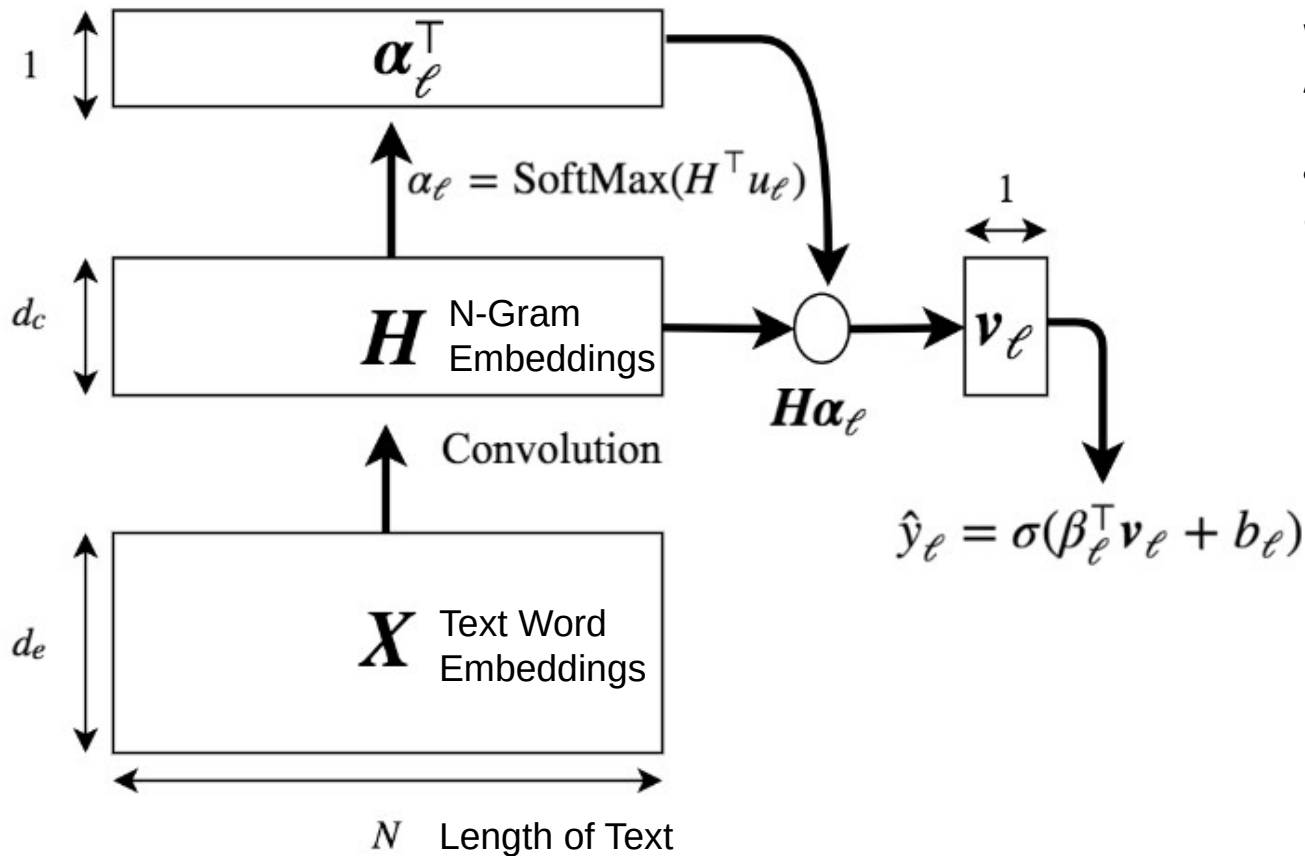
$$\alpha_\ell = \text{SoftMax}(H^\top u_\ell),$$

Weight representing
“match” between n-grams in document (H) and label embedding

$$v_\ell = \sum_{n=1}^N \alpha_{\ell,n} h_n.$$

N-grams weighted
by weights alpha
form document
vector v

Convolutional Attention Model



$$\alpha_\ell = \text{SoftMax}(H^\top u_\ell),$$

Weight representing
“match” between n-grams in document (H) and label embedding

$$v_\ell = \sum_{n=1}^N \alpha_{\ell,n} h_n.$$

N-grams weighted
by weights alpha
form document
vector v

Training Loss Function

$$L_{\text{BCE}}(\mathbf{X}, \mathbf{y}) = - \sum_{\ell=1}^{\mathcal{L}} y_{\ell} \log(\hat{y}_{\ell}) \\ + (1 - y_{\ell}) \log(1 - \hat{y}_{\ell}),$$

Binary Cross-Entropy Loss

- Binary classification for each possible label

$$L(\mathbf{X}, \mathbf{y}) = L_{\text{BCE}} + \lambda \frac{1}{n_y} \sum_{\ell: y_{\ell}=1}^{\mathcal{L}} \|\mathbf{z}_{\ell} - \boldsymbol{\beta}_{\ell}\|_2,$$

Regularize to make Labels
Representations Close to Code
Descriptions

Training Loss Function

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Regularize to make Labels
Representations Close to Code
Descriptions

Number of true labels
for data example

Representation of label based
on code description

Representation of
label learned by model

Dataset

- MIMIC 3
- Input: Sequence of words describing the stay of a patient in the ICU
(Max length of 2500 words)
- Output: ICD-9 codes describing diagnoses and procedures
- 9,000 unique ICD-9 codes:
 - 7,000 diagnosis codes and 2,000 treatment codes
- Preprocessing (100 dimension embeddings)

MIMIC-III full

# training documents	47,724
Vocabulary size	51,917
Mean # tokens per document	1,485
Mean # labels per document	15.9
Total # labels	8,922

Baselines

- Single Layer CNN
- Logistic Regression
- Bidirectional Gated Recurrent Unit

Evaluation Metrics

- Micro-averaged and Macro-averaged F1 and AUC
- Precision@n

(Most common n labels in the ground truth data)

$$\text{Micro-R} = \frac{\sum_{\ell=1}^{|\mathcal{L}|} \text{TP}_{\ell}}{\sum_{\ell=1}^{|\mathcal{L}|} \text{TP}_{\ell} + \text{FN}_{\ell}}$$

$$\text{Macro-R} = \frac{1}{|\mathcal{L}|} \sum_{\ell=1}^{|\mathcal{L}|} \frac{\text{TP}_{\ell}}{\text{TP}_{\ell} + \text{FN}_{\ell}},$$

Results

Model	AUC		F1				P@n	
	Macro	Micro	Macro	Micro	Diag	Proc	8	15
Scheurwegs et. al (2017)	–	–	–	–	0.428	0.555	–	–
Logistic Regression	0.561	0.937	0.011	0.272	0.242	0.398	0.542	0.411
CNN	0.806	0.969	0.042	0.419	0.402	0.491	0.581	0.443
Bi-GRU	0.822	0.971	0.038	0.417	0.393	0.514	0.585	0.445
CAML	0.895	0.986*	0.088	0.539*	0.524*	0.609*	0.709*	0.561*
DR-CAML	0.897	0.985	0.086	0.529	0.515	0.595	0.690	0.548

- The CAML model gives the strongest results on all metrics
- DR-CAML is the regularized version using $\lambda = 0.01$
- The Bi-GRU architecture is comparable to the vanilla CNN
- Logistic Regression worse than all neural architectures.

Interpretability Evaluation

- Sample of 100 predicted codes
- Using the most important k -gram ($k=4$) + five words on each side for context
- Asked the Physician how informative it is

934.1: "Foreign body in main bronchus"

CAML (HI)	<i>...line placed bronchoscopy performed showing large mucus plug on the left on transfer to...</i>
Cosine Sim	<i>...also needed medication to help your body maintain your blood pressure after receiving iv...</i>
CNN	<i>...found to have a large lll lingular pneumonia on chest x ray he was...</i>
Logistic Regression	<i>...impression confluent consolidation involving nearly the entire left lung with either broncho-centric or vascular...</i>

442.84: "Aneurysm of other visceral artery"

CAML (I)	<i>...and gelfoam embolization of right hepatic artery branch pseudoaneurysm coil embolization of the gastroduodenal...</i>
Cosine Sim	<i>...coil embolization of the gastroduodenal artery history of present illness the pt is a...</i>
CNN	<i>...foley for hemodynamic monitoring and serial hematocrits angio was performed and his gda was...</i>
Logistic Regression (I)	<i>...and gelfoam embolization of right hepatic artery branch pseudoaneurysm coil embolization of the gastroduodenal...</i>

428.20: "Systolic heart failure, unspecified"

CAML	<i>...no mitral valve prolapse moderate to severe mitral regurgitation is seen the tricuspid valve...</i>
Cosine Sim	<i>...is seen the estimated pulmonary artery systolic pressure is normal there is no pericardial...</i>
CNN	<i>...and suggested starting hydralazine imdur continue aspirin arg admitted at baseline cr appears patient...</i>
Logistic Regression (HI)	<i>...anticoagulation monitored on tele pump systolic dysfunction with ef of seen on recent echo...</i>

How to pick the most influential 4-gram for each label?

- **CAML** Looks at the SoftMax output α_{ℓ} to get the best 4-gram for a code prediction ℓ
- **Logistic regression** takes the sum of the coefficients of the weight matrix for ℓ , over the words in the k -gram. The top-scoring k -gram is then returned.
- **Code descriptions** calculate a word similarity (cosine similarity) metric between each stemmed k -gram and the stemmed ICD-9 code description.

- **Max-Pooling CNN**

$$a_i = \arg \max_{j \in \{1, \dots, m-k+1\}} (H_{ij}),$$

$$\alpha_{i\ell} = \sum_{j: a_j=i}^{d_c} \beta_{\ell,j}.$$

We then select the most important k -gram for a given label as $\arg \max_i \alpha_{i\ell}$.

Method	Informative	Highly informative
CAML	46	22
Code Descriptions	48	20
Logistic Regression	41	18
CNN	36	13

Table 7: Qualitative evaluation results. The columns show the number of examples (out of 100) for which each method was selected as “informative” or “highly informative”.

- CAML selects the greatest number of “highly informative” explanations, and selects more “informative” explanations than both the CNN baseline and the logistic regression model.
- Cosine Similarity metric also performs well the examples in Table 1 demonstrates the strengths of CAML.

Conclusion

Concluding Thoughts

- Yields strong improvements over previous metrics across several formulations of the ICD-9 code prediction task
- Provides satisfactory explanations for its predictions per stakeholder evaluation

Future Work

- Extensible
 - To other multi-label document tagging task
- Linguistic perspective
 - Integrate document structure of discharge summaries in MIMIC-III
 - Better handle non-standard writing and other sources of out-of-vocabulary tokens
- Application perspective
 - Build models that leverage hierarchy of ICD codes
 - Predict codes for future visits

Discussion

1. What are potential limitations of CAML?
2. The explanations provided from the attention mechanism are in the form of extracted snippets of text from the input document. Does this provide enough information? What else could be provided with this explanation?
3. Strengths and weaknesses of their evaluation and results (for both accuracy and interpretability)
4. What additional applications could CAML be useful for?

References

1. Mullenbach, James, et al. "Explainable prediction of medical codes from clinical text." arXiv preprint arXiv:1802.05695 (2018).
2. Scriven, Sarah. "An Introduction to Clinical Coding." Federation for Informatics Professionals (FEDIP), FEDIP, 14 Mar. 2022, <https://www.fedip.org/post/what-is-clinical-coding>.

Attention is not Explanation

Authors: Sarthak Jain, Byron C. Wallace

Presenters: Nikhil Nayak, Sree Harsha Tanneru, Hongjin Lin

2023/03/06

Attention Mechanism is a “recent” breakthrough in NLP

2015: birth of attention mechanism in NLP

- “Neural machine translation by jointly learning to align and translate”
Bahdanau et al. 2015

2017: birth of transformers based on self attention mechanism

- “Attention is all you need” Vaswani et al. 2017

Task: Hotel service

you get what you pay for . not the cleanest rooms but bed was clean and so was bathroom . bring your own towels though as very thin . **service was excellent** , let us book in at 8:30am ! for location and price , this ca n't be beaten , but it is cheap for a reason . if you come expecting the hilton , then book the hilton ! for uk travellers , think of a blackpool b&b.

Fig. 1. Example of attention visualization for an aspect-based sentiment analysis task, from [1, Fig. 6]. Words are highlighted according to attention scores. Phrases in bold are the words considered relevant for the task or human rationales.

Many use attention as an explanation mechanism

Show, Attend
Generate

Kelvin Xu
Jimmy Lei Ba
Ryan Kiros
Kyunghyun Cho
Aaron Courville
Ruslan Salakhutdinov
Richard S. Zemel
Yoshua Bengio

RETA
Healthcare

Chen et al. *BMC Medical Informatics and Decision Making* 2020, **20**(Suppl 3):131
<https://doi.org/10.1186/s12911-020-1110-7>

BMC Medical Informatics and
Decision Making

RESEARCH

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Interpretable clinical prediction via attention-based neural network



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Guangzhou, China. 22-24 November 2019

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But does attention necessarily provide faithful explanation?

What do you think?

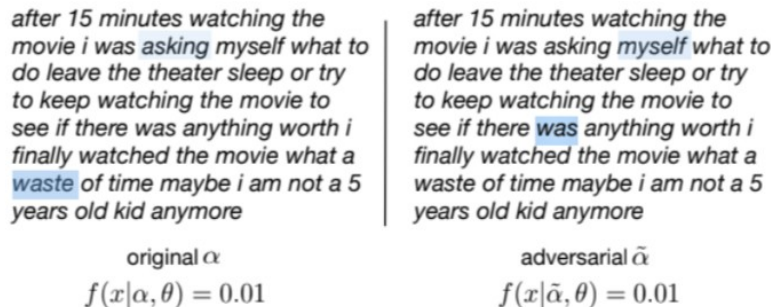


Figure 1: Heatmap of attention weights induced over a negative movie review. We show observed model attention (left) and an adversarially constructed set of attention weights (right). Despite being quite dissimilar, these both yield effectively the same prediction (0.01).

Research Questions

What is the degree to which attention weights provide meaningful “explanations” for predictions?

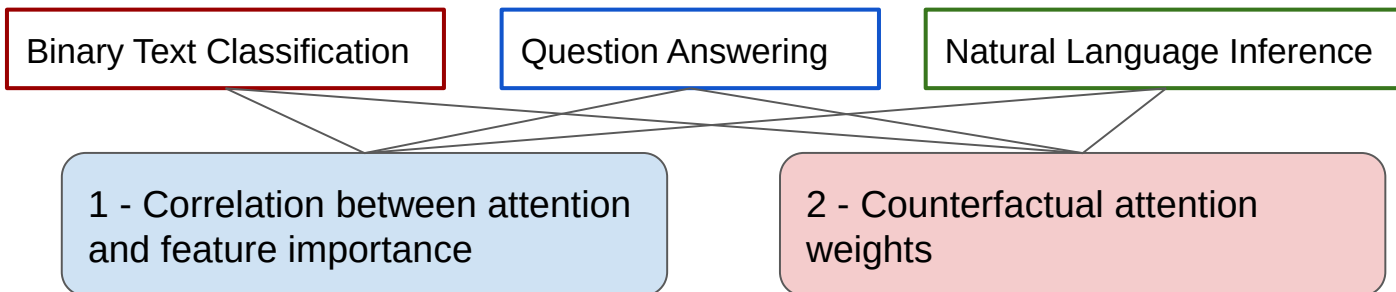
1. **Consistency:** To what extent do induced attention weights **correlate with measures of feature importance** – specifically, those resulting from gradients and leave-one-out (LOO) methods?
2. **Counterfactual attention weights:** Would **alternative attention weights** (and hence distinct heatmaps/“explanations”) necessarily yield different predictions?

Contributions and main findings

1. Standard attention modules **do not provide meaningful explanations** and should not be treated as though they do.
2. Learned attention weights are frequently **uncorrelated with gradient-based** measures of feature importance.
3. One can identify very **different attention distributions** that nonetheless yield equivalent predictions.

Experimental setup

<i>Dataset</i>	<i> V </i>	<i>Avg. length</i>	<i>Train size</i>	<i>Test size</i>	<i>Test performance (LSTM)</i>
SST	16175	19	3034 / 3321	863 / 862	0.81
IMDB	13916	179	12500 / 12500	2184 / 2172	0.88
ADR Tweets	8686	20	14446 / 1939	3636 / 487	0.61
20 Newsgroups	8853	115	716 / 710	151 / 183	0.94
AG News	14752	36	30000 / 30000	1900 / 1900	0.96
Diabetes (MIMIC)	22316	1858	6381 / 1353	1295 / 319	0.79
Anemia (MIMIC)	19743	2188	1847 / 3251	460 / 802	0.92
CNN	74790	761	380298	3198	0.64
bAbI (Task 1 / 2 / 3)	40	8 / 67 / 421	10000	1000	1.0 / 0.48 / 0.62
SNLI	20982	14	182764 / 183187 / 183416	3219 / 3237 / 3368	0.78



Correlation between Attention and Feature Importance

To what extent do induced attention weights correlate with measures of feature importance?

Specifically two methods are considered:

1. Feature gradient methods
2. Leave-One-Out (LOO) method

Gradient based Feature Importance

- To use the feature gradient method, we compute the gradient of the output with respect to each input feature.
- This tells us how much changing each feature would affect the output.
- We can visualize this by highlighting the input features that have the highest absolute gradient values.

Feature Erasure/LOO

- To use the LOO method, we remove one input feature at a time and observe how the output changes.
- This allows us to measure the impact of each feature on the output.
- We can visualize this by highlighting the input feature that is removed and showing the change in the output.

Distribution Change Measure for Experiments

Total Variation Distance (TVD), Kendall's τ coefficient

$$\text{TVD}(\hat{y}_1, \hat{y}_2) = \frac{1}{2} \sum_{i=1}^{|\mathcal{Y}|} |\hat{y}_{1i} - \hat{y}_{2i}|$$

$$\tau = \frac{(\text{number of concordant pairs}) - (\text{number of discordant pairs})}{(\text{number of pairs})} = 1 - \frac{2(\text{number of discordant pairs})}{\binom{n}{2}}$$

Algorithm for Feature Importance Computations

$$\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}), \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$$

$$\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \alpha)$$

$$g_t \leftarrow \left| \sum_{w=1}^{|V|} \mathbb{1}[\mathbf{x}_{tw} = 1] \frac{\partial y}{\partial \mathbf{x}_{tw}} \right|, \forall t \in [1, T]$$

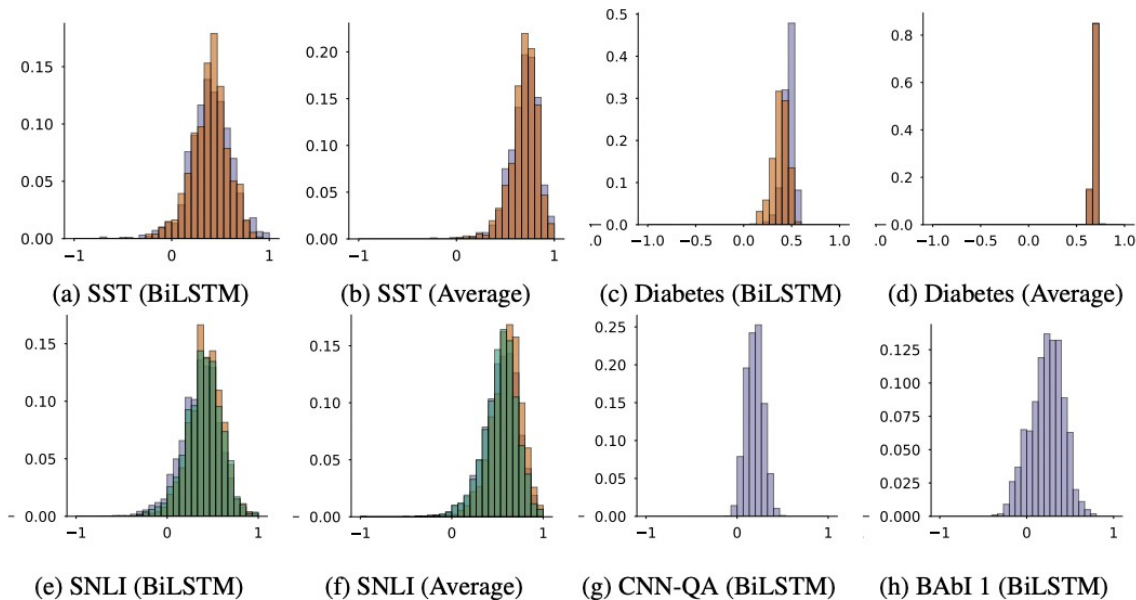
$$\tau_g \leftarrow \text{Kendall-}\tau(\alpha, g)$$

$$\Delta \hat{y}_t \leftarrow \text{TVD}(\hat{y}(\mathbf{x}_{-t}), \hat{y}(\mathbf{x})) , \forall t \in [1, T]$$

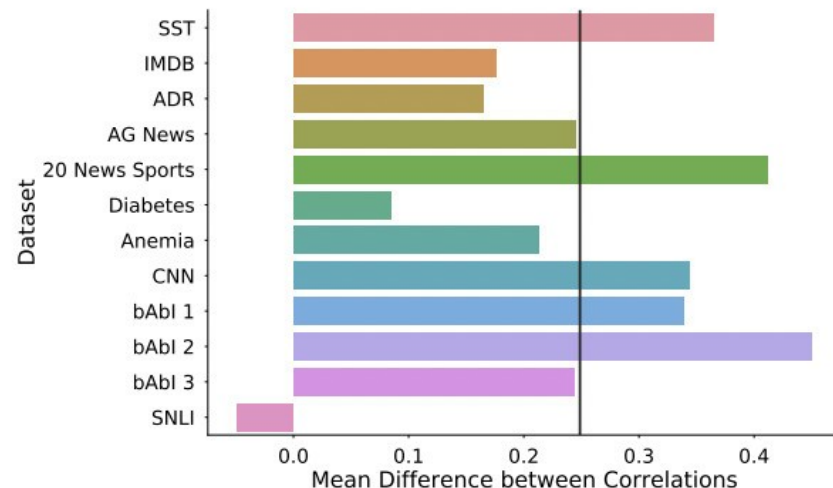
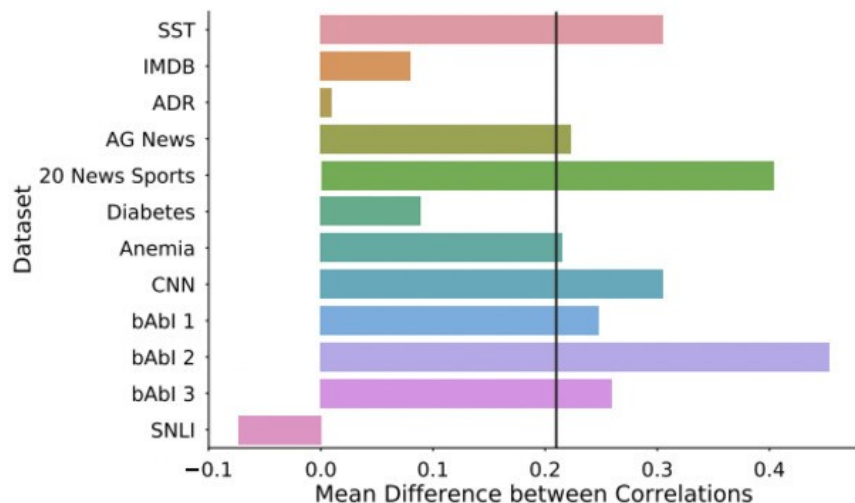
$$\tau_{loo} \leftarrow \text{Kendall-}\tau(\alpha, \Delta \hat{y})$$

Feature Gradients Results

Histogram of Kendall correlation between attention and gradients.



Correlation Comparison - Attention, Gradients, LOO



Mean difference in correlation.

Fig 1 - LOO, Gradients V/S Attention, LOO. Fig 2 - LOO, Gradients V/S Attention, Gradients.

Possible Limitations in Experiment Results

1. Potential issues with Kendall T correlation measure - Many irrelevant features may add noise to the correlation measures.
2. Alternative measures of feature importance, such as gradients and LOO, may not necessarily be considered ideal or deemed as the 'ground truth' for comparison purposes.

Counterfactual Attention Weights

What is the degree to which attention weights provide meaningful “explanations” for predictions?

1. **Consistency** i.e; correlation with other measures of feature importance
2. **Counterfactual attention distributions** should yield corresponding changes in predictions.

brilliant	and	moving	performances	by	tom	and	peter	finch
brilliant	and	moving	performances	by	tom	and	peter	finch
brilliant	and	moving	performances	by	tom	and	peter	finch

Counterfactual Attention Weights

Would the prediction be different if the model attended to different input features ?

Two experiments

1. **Attention Permutation:** Scramble the original attention weights
2. **Adversarial Attention:** Generate an adversarial distribution that is maximally distinct from the observed attention weights, whilst simultaneously close to the observed prediction

Attention Permutation - Algorithm

$$\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}), \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$$

$$\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \hat{\alpha})$$

for $p \leftarrow 1$ to 100 **do**

$$\alpha^p \leftarrow \text{Permute}(\hat{\alpha})$$

$$\hat{y}^p \leftarrow \text{Dec}(\mathbf{h}, \alpha^p) \quad \triangleright \text{Note : } \mathbf{h} \text{ is not changed}$$

$$\Delta \hat{y}^p \leftarrow \text{TVD}[\hat{y}^p, \hat{y}]$$

end for

$$\Delta \hat{y}^{med} \leftarrow \text{Median}_p(\Delta \hat{y}^p)$$

Attention Permutation - Results

Plot Δy at different peak attention values

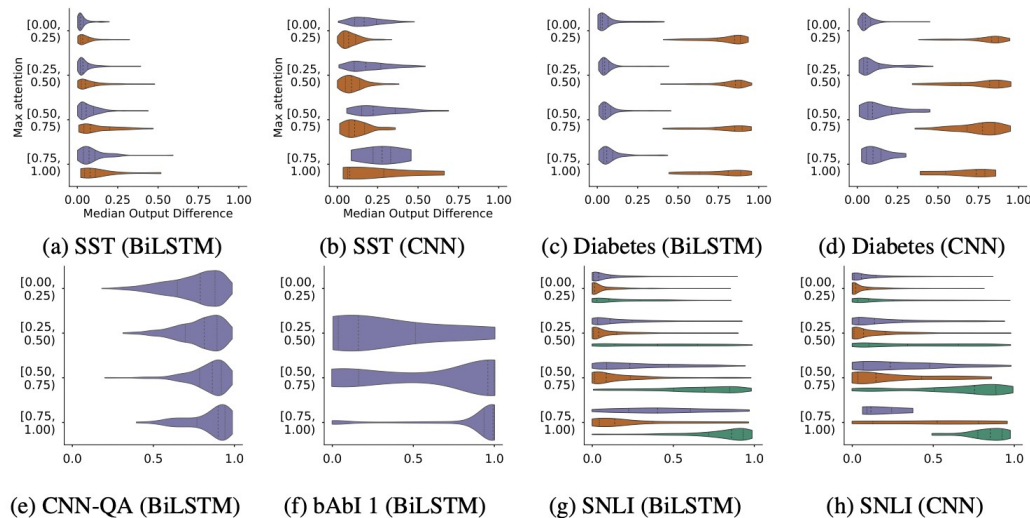


Figure 6: **Median change in output $\Delta \hat{y}^{med}$** (x-axis) densities in relation to the **max attention ($\max \hat{\alpha}$)** (y-axis) obtained by randomly permuting instance attention weights. Encoders denoted parenthetically. Plots for all corpora and using all encoders are available online.

Adversarial Attention

- Explicitly seek out attention weights that differ as much as possible from observed attention weights, yet leaving prediction almost unchanged.
- Why ?
- Alternative attention distributions identified for the same output may be viewed as equally plausible explanations for the same output. Is this undesirable ?
 - Cannot conclude that model made a specific prediction because it attended over inputs in a certain way.
 - Lack of specificity makes constructing counterfactual explanations hard ?
 - Makes model less trustworthy ?

Adversarial Attention - Optimisation

Seek $\alpha^{(1)}, \alpha^{(2)}, \alpha^{(3)} \dots \alpha^{(k)}$.

ϵ - Quantifies small difference in model output

Measure of distance between attention distributions ?

$$JSD(\alpha_1, \alpha_2) = \frac{1}{2} (KL[\alpha_1 | \frac{\alpha_1 + \alpha_2}{2}] + KL[\alpha_2 | \frac{\alpha_1 + \alpha_2}{2}])$$

$$\begin{aligned} & \underset{\alpha^{(1)}, \dots, \alpha^{(k)}}{\text{maximize}} && f(\{\alpha^{(i)}\}_{i=1}^k) \\ & \text{subject to} && \forall i \text{ TVD}[\hat{y}(\mathbf{x}, \alpha^{(i)}), \hat{y}(\mathbf{x}, \hat{\alpha})] \leq \epsilon \end{aligned} \quad (1)$$

Where $f(\{\alpha^{(i)}\}_{i=1}^k)$ is:

$$\sum_{i=1}^k JSD[\alpha^{(i)}, \hat{\alpha}] + \frac{1}{k(k-1)} \sum_{i < j} JSD[\alpha^{(i)}, \alpha^{(j)}] \quad (2)$$

Adversarial Attention - Algorithm

$\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}), \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$

$\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \hat{\alpha})$

$\alpha^{(1)}, \dots, \alpha^{(k)} \leftarrow \text{Optimize Eq 1}$

for $i \leftarrow 1$ to k **do**

$\hat{y}^{(i)} \leftarrow \text{Dec}(\mathbf{h}, \alpha^{(i)})$ $\triangleright \mathbf{h}$ is not changed

$\Delta \hat{y}^{(i)} \leftarrow \text{TVD}[\hat{y}, \hat{y}^{(i)}]$

$\Delta \alpha^{(i)} \leftarrow \text{JSD}[\hat{\alpha}, \alpha^{(i)}]$

end for

$\epsilon\text{-max JSD} \leftarrow \max_i \mathbb{1}[\Delta \hat{y}^{(i)} \leq \epsilon] \Delta \alpha^{(i)}$

Adversarial Attention - Results

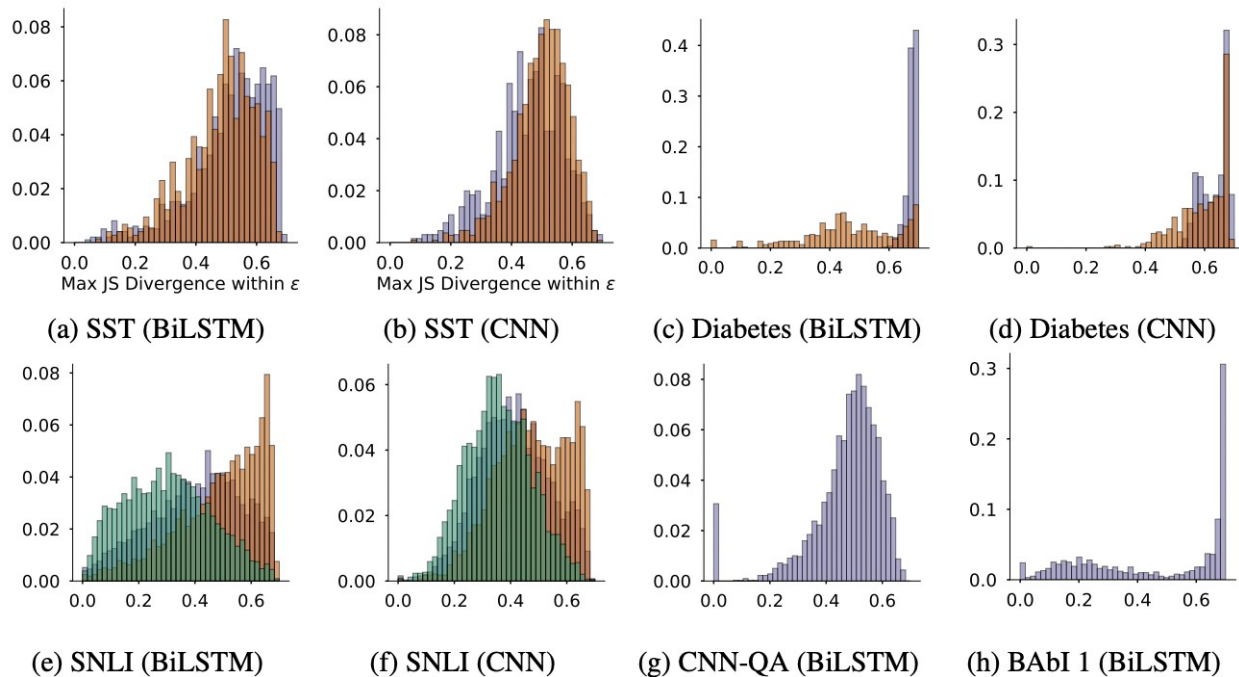


Figure 7: Histogram of **maximum adversarial JS Divergence (ϵ -max JSD)** between original and adversarial attentions over all instances. In all cases shown, $|\hat{y}^{adv} - \hat{y}| < \epsilon$. Encoders are specified in parantheses.

Adversarial Attention - Results

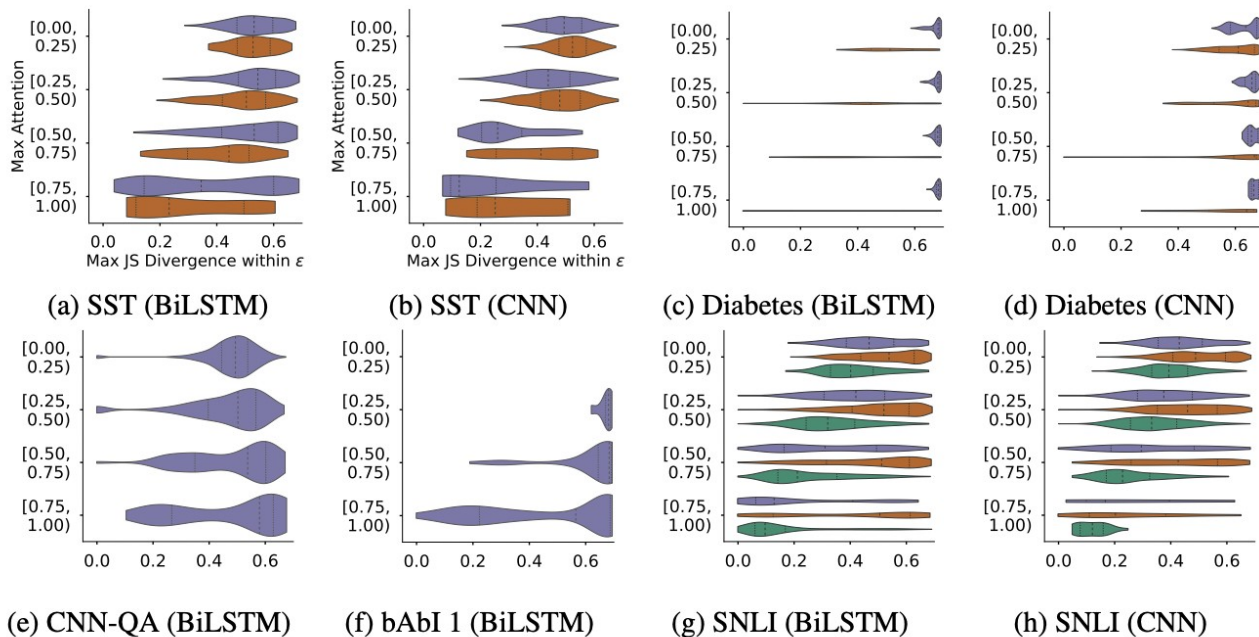


Figure 8: Densities of **maximum JS divergences** (ϵ -**max JSD**) (x-axis) as a function of the **max attention** (y-axis) in each instance for obtained between original and adversarial attention weights.

Counterfactual Attention - Limitations

- Adversarial weights may themselves be unlikely under model parameters
- There are instances where a single explanation out of multiple plausible explanations is sufficient.

Overall Limitations

- Alternate attention mechanisms are not explored
- Limited evaluations to classification, QA, and NPI tasks. seq2seq tasks are not considered.

Critique and Discussion

- Wiegrefe, S. & Pinter, Y. (2019). Attention is not not Explanation. In Proceedings of the 2019 Conference on EMNLP-IJCNLP (pp. 11-20).
 - Attention distribution is not a primitive.
 - Is attention necessary for prediction ?
 - Is the variance in attention distributions unusual ? i.e; How adversarial are the adversaries ?
- Do you agree with the authors' conditions of a "faithful explanation"?
- Do you think attention constitutes a "sufficient" explanation ?
- What other conditions should we consider as a good explanation?
- How can experts be brought into evaluate conditions of good explanations?